fit_sjsdm_cross_validation_candidate.R

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`fit_sjsdm_cross_validation_candidate`
Fit an sjSDM Cross-Validation Candidate to Convergence

Description

Fits one candidate-fold from its initial CV iteration budget and retries a successful non-converged fit with doubled iterations up to the exact maximum.

Usage

```
fit_sjsdm_cross_validation_candidate(
  data_train_input = NULL,
  candidate = NULL,
  sel_abiotic_formula = NULL,
  config_sjsdm_cv_fitting = NULL,
  seed = 900723L,
  repeat_id = NA_integer_,
  fold_id = NA_integer_,
  device = "cpu",
  biotic = NULL,
  fit_function = fit_sjsdm_regularization_candidate,
  convergence_function = diagnose_jsdm_convergence
)
```

Arguments

`data_train_input`, `candidate`, `sel_abiotic_formula`
: Candidate fitting inputs passed to [fit_sjsdm_regularization_candidate()].

`config_sjsdm_cv_fitting`
: Validated CV fitting configuration containing initial and maximum budgets.

`seed`
: Deterministic seed reused unchanged for every attempt.

`repeat_id`, `fold_id`
: Integer CV identifiers persisted in attempt provenance.

`device`, `biotic`
: Fitting device and optional biotic structure passed to the candidate fitter.

`fit_function`, `convergence_function`
: Injectable fitting and convergence functions.

Value

An 'sjsdm_cv_fit_result' list containing the last model, final status, convergence fields, actual budget, and an attempt-level table. Failed fits retain the same scalar diagnostic fields as successful fits, using typed missing values where no convergence history is available.

Examples

```
## Not run:
fit_sjsdm_cross_validation_candidate(
  data_train_input = data_train_input,
  candidate = data_candidate,
  sel_abiotic_formula = formula_jsdm_environment,
  config_sjsdm_cv_fitting = config_sjsdm_cv_fitting,
  repeat_id = 1L,
  fold_id = 1L
)

## End(Not run)
```

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